(1) A sequence is generated by the formula $M_n = an^2 + bn + c$, where a, b and c are constants. If $M_1 = 4$, $M_2 = 10$ and $M_3 = 18$, find the values of a, b and c.

Solution : In $M_n = an^2 + bn + c$, we take $n = 1, 2, 3$ and evaluate the expression. We get

$1a + 1b + c = 4$

$4a + 2b + c = 10$

$9a + 3b + c = 18$

Solving these equations we get values of a,b,c. (write your calculations)

$a+b+c=4$ — (I)
$4a+2b+c=10$ — (II)
$9a+3b+c=18$ — (III)
Equ^n (II) − Equ^n (I)
$4a+2b+c=10$
$-a+b+c=4$
$3a+b=6$ — (IV)
Equ^n (III) − Equ^n (II)
$9a+3b+c=18$
$4a+2b+c=10$
$5a+b=8$ — (V)
Equ^n (V) − Equ^n (IV)

$5a+b=8$
$-3a+b=6$
$2a=2$
$\therefore a=1$
Put $a=1$ in equ^n (IV)
$3+b=6$
$\therefore b=3$
Put $a=1, b=3$ in equ^n (I)
$1+3+c=4$
$\therefore c=4-4$
$\therefore c=0$
$a=1, b=3, c=0$

2) Evaluate : (a) $\sum_{n=1}^{6} \frac{2}{5}(3)^n$ (b) $729-243+81-\ldots\ldots\ldots\ldots \oplus 3 \ominus 1^{+}$ (correction)

(c) $\sum_{n=1}^{\infty} 8(0.25)^n$

Solution : (a) $\sum_{n=1}^{6} \frac{2}{5}(3)^n = \frac{2}{5}\sum_{n=1}^{6}(3)^n = \frac{2}{5}[3+3^2+3^3+3^4+3^5+3^6]$

$= \frac{2}{5}\left[\frac{3(3^6-1)}{3-1}\right]$

$= \frac{2184}{5}$

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(b) $729-243+81-\ldots\ldots\oplus 3\ominus 1 = 729-243+81-\ldots-3+1$

Here $a = 729$, $r = \frac{1}{3}$, $t_n = 1$

$t_n = ar^{n-1}$

$1 = 729\left(-\frac{1}{3}\right)^{n-1}$

$\frac{1}{729} = \left(-\frac{1}{3}\right)^{n}\left(-\frac{1}{3}\right)^{-1}$

$\frac{1}{729} = \left(-\frac{1}{3}\right)^{n} \times \frac{1}{\left(-\frac{1}{3}\right)^{1}}$

$\frac{1}{3^6} \times -\frac{1}{3} = \left(-\frac{1}{3}\right)^{n}$

$\left(-\frac{1}{3}\right)^{7} = \left(-\frac{1}{3}\right)^{n}$

$\therefore n = 7$

Now $a = 729$, $r = \frac{1}{3}$, $n = 7$

$S_n = a\left(\frac{1-r^n}{1-r}\right) = 729\left(\frac{1-\left(-\frac{1}{3}\right)}{1-\left(-\frac{1}{3}\right)}\right) = 547$

(c) $\sum_{n=1}^{\infty} 8(0.25)^n = 8\sum_{n=1}^{\infty}(0.25)^n = 8\sum_{n=1}^{\infty}\left(\frac{1}{4}\right)^n = 8\left[\frac{1}{4}+\left(\frac{1}{4}\right)^2+\left(\frac{1}{4}\right)^3+\ldots\right]$

This is a G.P with $a = \frac{1}{4}$ and $r = 1/4$

Since $|r| = |1/4| < 1$, the sum to infinity of this G.P exist

$S = \frac{a}{1-r} = \frac{1/4}{1-1/4} = \frac{1/4}{3/4} = \frac{1}{3}$

$\sum_{n=1}^{\infty} 8(0.25)^n = 8 \times \frac{1}{3} = \frac{8}{3}$

3) If the ratio of H.M. and G.M. of two numbers is 12 : 13, then show that the ratio of the numbers is 9 : 4.

Solution : Let numbers be a and b.

$H.M. = \frac{2ab}{a+b}$ and $G.M = \sqrt{ab}$

The ratio of H.M. and G.M. of a and b is 12 : 13.

$\therefore \frac{HM}{GM} = \frac{12}{13}$

$\therefore \frac{2ab}{a+b} \div \sqrt{ab} = \frac{12}{13}$

$\therefore \frac{2ab}{a+b} \times \frac{1}{\sqrt{ab}} = \frac{12}{13}$

$\therefore$

$\therefore \frac{\sqrt{ab}}{a+b} = \frac{6}{13}$

$\therefore$ Squaring both sides

$\therefore \frac{ab}{(a+b)^2} = \frac{36}{169}$

$\therefore 36(a^2+2ab+b^2) = 169ab$

$\therefore 36a^2 + 72ab + 36b^2 = 169ab$

$\therefore 36a^2 - 97ab + 36b^2 = 0$

$36a^2 - 81ab - 16ab + 36b^2 = 0$

$9a(4a-9b) - 4b(4a-9b) = 0$

$(4a-9b)(9a-4b) = 0$

$4a-9b = 0,\ 9a-4b = 0$

$\therefore 4a = 9b,\ 9a = 4b$

$\therefore \frac{a}{b} = \frac{9}{4}$ or $\frac{a}{b} = \frac{4}{9}$

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$\therefore$ .....

$\therefore a : b = 9 : 4$

4) If a, b, c are in H.P., then show that $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in H.P.

Solution : Given a, b, c are in H.P.

$\therefore b = \frac{2ac}{a+c}$

$\therefore b(a+c) = 2ac$ ---- (1)

$\therefore ab + bc = 2ac$ ..... (1)

To show that $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in H.P.

i.e. to show that $\frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ are in A.P.

i.e. to show that $2\left(\frac{c+a}{b}\right) = \frac{b+c}{a} + \frac{a+b}{c}$

R.H.S $= \frac{b+c}{a} + \frac{a+b}{c}$

$= \frac{bc+c^2+a^2+ab}{ac}$

$= \frac{c^2+a^2+ab+bc}{ac}$

$= \frac{c^2+a^2+2ac}{ac}$ [by (2)]

$= \frac{(c+a)^2}{ac}$

$= \frac{b(c+a)(c+a)}{bac}$

$= \frac{2ac\,(c+a)}{bac}$ [by (1)]

$= 2\left(\frac{c+a}{b}\right)$

$=$ L.H.S (Proved)

5) The sum of three decreasing numbers in A.P. is 27. If -1, -1,3 are added to them respectively, the resulting series is a G.P. Find the numbers.

Solution : Let three decreasing numbers in A.P. be $a+d, a, a-d$

Their Sum is 27

$(a+d) + a + (a-d) = 27$

$\therefore 3a = 27 \Rightarrow \boxed{a = 9}$

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Now adding $-1, -1$ and 3 to the three numbers we get $a+d-1, a-1, a-d+3$

Now it is given that these resulting numbers are in G.

$\therefore \frac{a+d-1}{a-1} = \frac{a-1}{a-d+3}$

But $a=9$, $\frac{8+d}{8} = \frac{8}{12-d}$

$\therefore (8+d)(12-d) = 64$

$\therefore 96 - 8d + 12d - d^2 = 64$

$\therefore d^2 - 4d - 32 = 0$

$\therefore d^2 - 8d + 4d - 32 = 0$

$\therefore d(d-8) + 4(d-8) = 0$

$\therefore (d-8)(d+4) = 0$

$\therefore d-8 = 0, d+4 = 0$

$\therefore d = 8$ or $d = -4$

Taking $a=9$, $d=8$ the three required no. are $a+d = 9+8 = 17$, $a = 9$, $a-d = 9-8 = 1$ i.e. $17, 9, 1$

These numbers are decreasing and are in A.P

6) **Find the sum of all three digit numbers which are not divisible by 7.**

Solution : Sum of all three digit numbers which are not divisible by 7 = Sum of all three digit numbers − Sum of all three digit numbers which are divisible by 7 —— (1)

Sum of all three digit numbers

$= 100 + 101 + 102 + \cdots + 999$ (The Series in A.P)

$= \frac{n}{2}[2a + (n-1)d]$ where $n = 900$, $a = 100$, $d = 1$

$= \frac{900}{2}[2(100) + 899 \times 1]$

$= 450 \times 1099$

$= 494550$ —— (2)

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Sum of all three digit numbers which are divisible by 7 = 105 + 112 + ---- + 994

Let 994 be nth term of the given series.

$\therefore 994 = 105 + (n-1) \times 7$ $\quad (\because t_n = a + (n-1)d)$

$\therefore 994 - 105 = (n-1) \times 7$

$\therefore 889 = 7n - 7$

$\therefore 889 + 7 = 7n$

$\therefore 7n = 896$

$\therefore n = \frac{896}{7}$

$\therefore \boxed{n = 128}$

$\therefore$ Sum of all three digit numbers which are divisible by 7 $= \frac{n}{2}[2a + (n-1)d]$

$= \frac{128}{2}[2 \times 105 + (128-1) \times 7]$

$= 64(210 + 127 \times 7)$

$= 64 \times (210 + 889)$

$= 64 \times 1099$

$= 70336$ ——(3)

From (1), (2) and (3)

Sum of all three digit numbers which are not divisible by 7 = 494550 − 70336

= 424214

Sign of Teacher :

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